

Riyasat Zaman

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Final-year Computing Science student (Statistics minor) and current software research assistant. Ships production full-stack systems end-to-end across React, Python, SQL, and cloud, with strong object-oriented fundamentals and a track record of picking up unfamiliar stacks and tools quickly. Comfortable working in Agile teams with Git and code review. Available immediately, authorized to work in Canada with no sponsorship required, open to relocation.

EXPERIENCE

Software Research Assistant — University of Alberta

Jun 2026 – Present

- Developing the UCommons App, a cross-platform Flutter mobile app for a five-year NSERC-funded smart building research platform, delivering indoor wayfinding and an augmented reality navigation view that overlays directional guidance on a live camera feed.
- Designing the REST API contract connecting the app to a Neo4j building digital twin, where path planning runs over a graph derived from IFC building data modeled with the BOT ontology.
- Building a modular, extensible architecture to support a multi-version data-collection rollout across a two-year research study under an approved ethics protocol.

Software Developer — Space Exploration Alberta Robotics (SPEAR)

Sep – Dec 2023

- Integrated SLAM workflows synchronizing LiDAR and IMU data for rover localization across 7+ navigation tests, reducing localization error by ~35% over the previous GPS-only approach.
- Built computer vision pipelines using OpenCV and PyTorch to automate terrain classification and real-time obstacle detection.
- Implemented real-time path planning and sensor fusion in Python and C++ with ROS, contributing to the full autonomous navigation stack used in live field tests.

Product Developer — UofA Blueprint

May – Dec 2022

- Shipped a production React/TypeScript/Tailwind frontend for ISA's ICard rewards app serving 10,000+ international students, implementing 10+ feature screens for vendor verification and discount redemption.
- Developed RESTful APIs with Node.js, Express.js, and PostgreSQL for user registration, content management, and admin workflows.
- Led QA and usability testing across 3 user evaluation rounds, improving Lighthouse performance score from ~65 to 90+.

PROJECTS

Verdigris — Distributed Social Network | Django, React, PostgreSQL, Docker | [Video Demo](#)

Apr 2025

- Built a Django/React distributed social network across 5 independently hosted nodes, enabling real-time cross-server posts, comments, likes, and follows between users on different servers.
- Designed REST API contracts for cross-node synchronization of authors, posts, and follows, ensuring consistent state across nodes.
- Deployed 5 cloud-hosted nodes with Docker, NGINX, and PostgreSQL; secured node-to-node requests with HTTP Basic Authentication.

TTC Transit Reliability Monitor | Python, Airflow, Snowflake, dbt, Docker, Streamlit | [GitHub](#) | [Demo](#)

May 2026

- Built an end-to-end Python data pipeline ingesting live TTC vehicle-location JSON every 15 minutes, landing raw data in Snowflake and transforming through dbt staging, intermediate, and mart layers.
- Orchestrated ingestion and transformation workflows with Apache Airflow and Docker Compose, processing 73,924+ vehicle observations across 218+ routes and 3,235+ distinct vehicles.
- Implemented 44 automated dbt and pytest checks validating source keys, timestamps, route relationships, and data quality contracts.

Capto — AI Image Captioning Tool | Flask, Hugging Face, GitHub Actions | [GitHub](#) | [Demo](#)

Mar 2025

- Built an AI captioning app using Flask and Qwen3-VL via Hugging Face Inference Providers, supporting image upload, caption generation, regeneration, editing, and one-click download.
- Deployed on Render with Gunicorn, environment-based configuration, fallback handling, and a GitHub Actions CI/CD pipeline.

EDUCATION

University of Alberta — BSc Computing Science, Statistics Minor

Completed Aug 2026 (Convocation Nov 2026)

Awards: University of Alberta International Country Scholarship; International Student Scholarship

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, R

Frameworks & Data: React, Flutter, Flask, Django, Node.js, Express.js, REST APIs, PostgreSQL, MySQL, MongoDB, Neo4j, Snowflake, Airflow, dbt

AI / DevOps / Cloud: Git, GitHub, CI/CD, Docker, AWS, Linux, Agile/Scrum, PyTorch, OpenCV, Hugging Face